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GROUP SPECIFICATION

Fifth Generation Fixed Network (F5G); F5G-A Architecture of Computing Collaboration in PON Network

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This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Fifth Generation Fixed Network (F5G).

Modal verbs terminology

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1 Scope

The present document specifies the architecture framework and requirements of computing collaboration in PON optical access networks and key computing functions and capabilities of PON network components.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI GS F5G 036 (V1.1.1) (2026-04): "Fifth Generation Fixed Network (F5G); F5G Advanced Network Architecture Release 4".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI GR F5G 008 (V1.1.1) (2024-06): "Fifth Generation Fixed Network (F5G); F5G Use Cases Release #2".

[i.2] ETSI GR F5G 020 (V1.1.1) (2024-10): "Fifth Generation Fixed Network (F5G); F5G Advanced Use Cases; Release 3".

[i.3] Recommendation ITU-T G.988 (2022) including Amendments: "ONU management and control interface (OMCI) specification".

[i.4] Recommendation ITU-T G.9943 (10/2025): "High speed fibre-based in-premises transceivers network management".

[i.5] Recommendation ITU-T G.9949 (10/2025): "Wireless local area network management control interface (WMCI) for in-premises network".

[i.6] BBF TR-069: "CPE WAN Management Protocol (CWMP)".

[i.7] BBF TR-181: "Device Data Model for TR-069".

[i.8] BBF TR-369: "User Services Platform (USP)".

[i.9] BBF TR-383: "Common YANG Modules for Access Networks".

[i.10] BBF TR-385: "ITU-T PON YANG Modules".

3 Definition of terms, symbols and abbreviations

3.1 Terms

Void.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

3D	Three Dimensions
6G	6th generation mobile networks
AI	Artificial Intelligence
AP	Access Point
CDN	Content Delivery Network
CO	Central Office
CPN	Customer Premises Network
CPU	Central Processing Unit
CWMP	CPE WAN Management Protocol
DBA	Dynamic Bandwidth Allocation
E2E	End-to-End
E-ONU	Edge ONU
FMCI	Fibre Management Control Interface
FTTR	Fibre-To-The-Room
GEM	GPON Encapsulation Method
GPU	Graphics Processing Unit
ICT	Information and Communications Technology
IoT	Internet of Things
IP	Internet Protocol
KPI	Key Performance Indicator
M&C	Management and Control
MCA	Management, Control & Analytics
MFU	Main Fibre Unit

NOTE: The MFU was called Primary ONU (P-ONU) in previous releases of the F5G and F5G-A architecture.

mTLS	mutual Transport Layer Security
NaaS	Network as a Service
NE	Network Element
NMP	Network Management Platform
NMS	Network Monitoring System
NPU	Neural Processing Unit
O&M	Operations and Maintenance
ODN	Optical Distribution Network
OLT	Optical Line Termination
OMCI	Optical Network Terminal Management & Control Interface
ONT	Optical Network Terminal

NOTE: An ONT is many times also referred to in the industry as an Optical Network Unit (ONU).

ONU	Optical Network Unit
OTN	Optical Transport Network
PON	Passive Optical Network
P-ONU	Primary ONU
QoD	Quality on Demand

QoE	Quality of Experience
QoS	Quality of Service
RBAC	Role-Based Access Control
SAP	Service Access Point
SFU	Sub Fibre Unit

NOTE: The SFU used to be called Edge ONU (E-ONU) in previous release of the F5G and F5G-A architecture.

SLA	Service Level Agreement
SMP	Service Mapping Point
SNR	Signal-to-Noise Ratio
SPP	Service Processing Point
TLS	Transport Layer Security
VLAN	Virtual Local Area Network
VoIP	Voice over Internet Protocol
VR	Virtual Reality
WAN	Wide Area Network
WLAN	Wireless Local Area Network
WMCI	Wi-Fi® Management & Control Interface
XR	eXtended Reality
YANG	Yet Another Next Generation

4 Overview of Computing Collaboration Architecture

4.1 Relationship with F5G-A Architecture

The F5G-A network architecture consists of three basic planes from F5G, an Underlay Plane, a Service Plane and a Management, Control and Analytics (F5G-A MCA) Plane. A cross-plane computing is specified as main characteristic in F5G-A network architecture. The compute-oriented resources are located on any of the network nodes or are co-located with the network nodes.

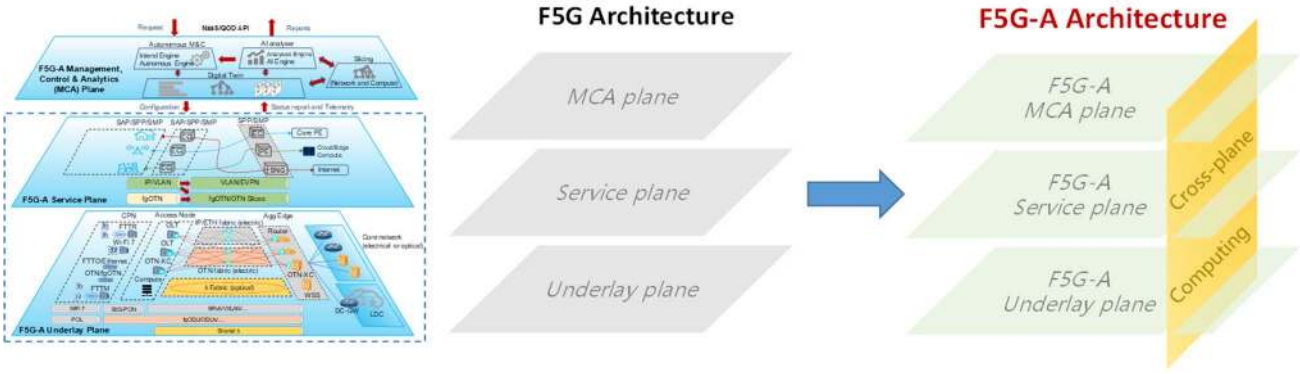


Figure 1: F5G-Advanced network architecture

As defined in the F5G-A network architecture [1], computing collaboration in the PON network requires the following functionalities:

- Compute resources are distributed throughout the network running on network elements or dedicated compute hardware.
- Computing collaborating with the network, enables the use of more intelligent algorithms including AI-based functions to steer and control the network. The network requires coordinated configuration of computing, storage, and networking. Computing and storage shall be addressed from the physical hardware (general purpose, AI-optimized and special purpose), cloud computing platform, and software packages.

- Support of a Northbound Interface to configure Network as a Service (NaaS) and to enable the establishment of connections with a certain Quality on Demand (QoD). This supports the real-time dynamic scheduling of computing power.
- Enabling operators to control and manage the Customer Premise Network (CPN) requires the capability to manage the services on the CPN segment. Computing resources in CPN and computing resource requirement of users are mastered by the service provider or network operator.

In the F5G-A network, the different planes should support the following features:

1) F5G-A Underlay Plane in the access and CPN network segment: the physical network plane, including all physical network nodes and devices. In PON/FTTR networks, F5G-A Underlay Plane includes OLT, ONU, MFU and SFU network nodes, which, shall support:
   - Computing capability.
   - Guaranteed communication capability.
   - Storage capacity.
   - Analytical capability.
   - Data collecting capacity.
   - An software operating system with standardized interfaces, where applications are installed.
2) F5G-A Service Plane: An Access Network Node typically contains SAP, SPP and SMP. It identifies services, adds or removes encapsulations, and directs the computing-related traffic to the proper underlay fabric, channels, and compute resources.
3) F5G-A Management, Control and Analytics (MCA) Plane: The MCA Plane serves as the intelligent core of the network, performing management, control and analytics of the complete F5G-A network. It is comprised of four logical components: Digital Twin, Autonomous Management and Control, AI Analyser and Slicing. For the PON/FTTR computation collaboration architecture, this plane should support:
   - Strong computing capability.
   - Strong storage capacity.
   - Strong analytical capability based on AI and Digital Twin.
   - Computing resource managements.

4.2 Architecture of computing collaboration

4.2.1 Architecture framework of computing collaboration

As shown in Figure 2, the main characteristic of the computing collaboration architecture is the distributed computing power across different devices in the optical access and CPN network, including the Network Management Platform (NMP), OLT, ONT(MFU) and SFU. For the collaboration two options are foreseen. First option is extending the existing management and control protocols with computing-collaboration-specific data models, so that the procedures defined in clause 4.2.2 are realized. Second option is establishing an IP-based computing collaboration data channel for high-frequency telemetry data exchange and AI model distribution, when the capacity or the latency of the management channel is not sufficient, the IP tunnel in Figure 2 is visualizing this option.

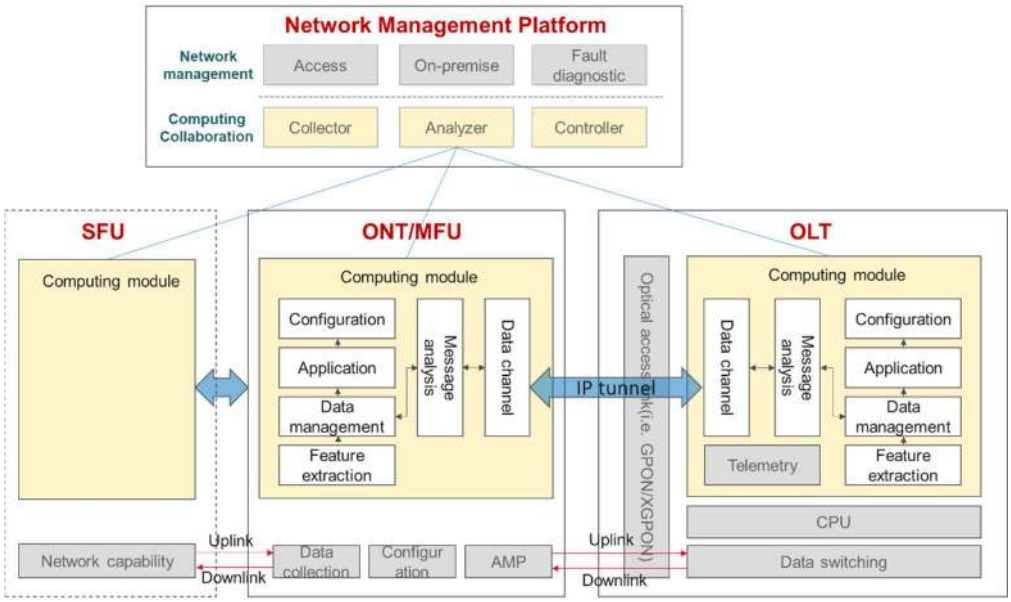


Figure 2: Architecture framework of computing collaboration

The main components achieving computing collaboration include:

1) Computing collaboration module: enables computing collaboration message exchange and processing.

2) Application module: processes and analyses data and makes collaboration decisions.

3) Data management: manages the data lifecycle and enables data subscription.

4) Configuration module: based on the decision, it generates configuration data for the devices in the optical access network.

The overall architecture of the convergence technology between the access network and the broadband customer network computing network is shown in Figure 3. According to functional divisions, it is composed of the "Forwarding Layer", "Computing Layer", "Management and Control Layer", and "Orchestration Layer".

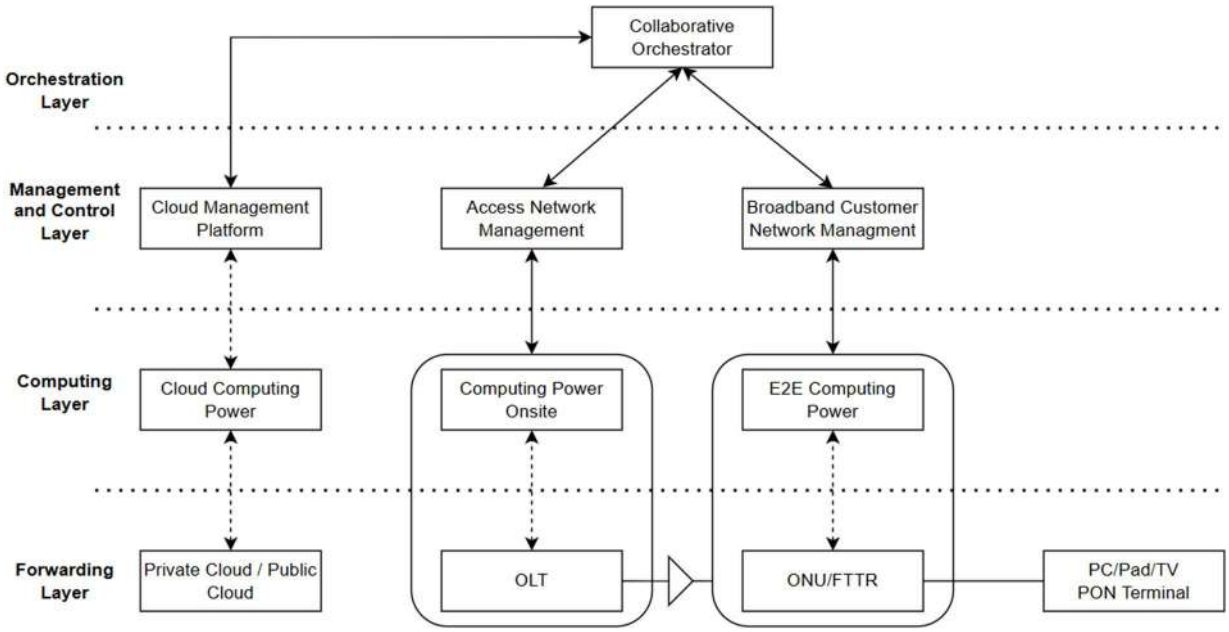


Figure 3: Schematic diagram of the overall architecture of the convergence technology between the access network and the broadband customer network computing network

The Orchestration Layer orchestrates functions of the management and control system and the cloud management system.

EXAMPLE: Collaboratively invoking intra-network computing power and off-network computing power, end-to-end collaboration between an access network controller and another network controller, and collaboration between intra-network edge enhanced computing power and access network connections are implemented.

The Management and Control Layer includes the cloud management platform, the access network management system, and the home network management system. The access network management system manages, configures, and maintains the access network NEs, service pipes, and computing power through the controller's southbound control interface and a standard information model. It also perceives, calculates, and analyses the status of the connected service through data interfaces. The access network management system accepts the unified scheduling of the orchestration and collaboration system through the northbound interface. The home network management system manages the FTTR MFU and SFU devices, manages, allocates, and maintains the FTTR devices, and coordinates and schedules the computing power between the internal computing power modules.

The Computing Layer includes the computing power within the access network (both, inside and outside the network), as well as the computing power at the device-side. The computing power within the network includes the local computing power at the OLT device and the device-side computing power at the home network element. The computing power outside the network includes the cloud computing power at edge clouds, central clouds, and public clouds. The Computing Layer provides perception, measurement, scheduling, and management of computing power. The computing power within the access network is opened to the users accessing the edge computing power as a service or product. Or it serves as edge computing centre for home or an industry usage. It can be used for running home network applications and vertical industry applications, enabling real-time, secure, and reliable lightweight computing.

The Forwarding Layer includes access network elements, ODN, and service pipelines. The access network elements include OLT devices, and home network elements (ONT/FTTR devices). The service pipelines include OLT user-side application-level service pipelines, home network user-side application-level service pipelines, and OLT network-side service pipelines with different SLA levels. The Forwarding Layer provides access, forwarding, and aggregation of user terminal service data, and provides application-level SLA deterministic assurance for different application data.

The entire access network and broadband customer network need to integrate multiple ICT capabilities, such as sensing, computing, and storage, to implement computing-network collaboration. The specific list of capabilities is as follows:

- **Awareness**: real-time awareness of user experience is supported for application-level service flows. Supports the awareness of the device status based on the device environment.
- **Calculation**: For service flows, the system collects and analyses real-time user data and evaluates user experience based on user usage. Accurately allocate network resources based on resource usage. The training and use of AI/ML models should be supported.
- **Connection**: application-level slice connections should be supported. Real-time connection parameters should be adjusted.
- **Storage**: it should support the storage of perception results and real-time data. It should also support the storage of data related to computation results. It should further support mechanisms for data synchronization and reporting, to ensure real-time closed-loop control and management functionality.
- **Control**: it should support the ability to identify specific events, such as poor-service quality identification. It should also support dynamic adjustments to service flows and networks based on real-time computation results. Additionally, it should support the allocation and control of resources such as computation and storage.

At the same time, the service processing capability of the entire network is achieved through independent closed-loop and cross-layer coordination at various levels, including the network management level, the central office equipment, and the broadband customer network.

4.2.2 General procedure of computing coordination

To implement a computing coordination procedure, different devices in the optical access network have dedicated roles. Figure 4 shows a general procedure of computing coordination:

1) Step 1: Enable function for dedicated ONT(MFU). Computing collaboration is a network service offered to end users by the service provider. The enabling of the function is triggered by the NMP and configured by the service provider. The targeted ONT(MFU) shall be informed via the OLT.

2) Step 2: The OLT informs the corresponding ONT(MFU) to enable the computing collaboration function triggering the negotiation of functional capabilities.

3) Step 3: The corresponding ONT(MFU) exchanges messages with the OLT via the management channel using OMCI to discover computing collaboration capabilities, negotiate the profile and create a data channel for computing collaboration.

4) Step 4: Based on collaboration function, the OLT subscribes the necessary data from ONT(MFU) for future decision making.

5) Step 5: the ONT(MFU) collects data from the on-premises network, performs necessary data filtering and characteristic extraction and sends the processed data to the OLT. Based on the received data, the OLT makes decisions and provides the strategy for network configuration

6) Step 6: The OLT forwards the decision to ONT(MFU) for dedicated services.

EXAMPLE: Decisions include new network configurations and examples for dedicated services are network optimization services.

7) Step 7: The OLT performs checks the outcome and provides a report to the NMP periodically.

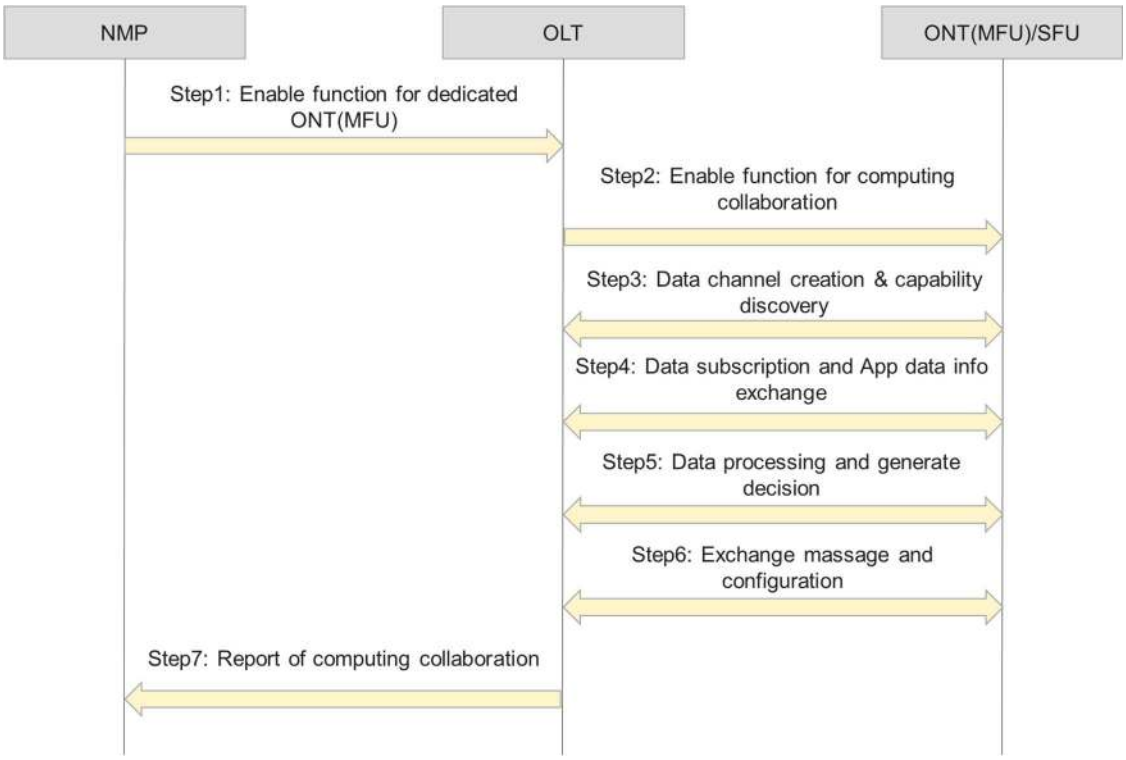


Figure 4: General procedure of computing coordination

NOTE: The management of computing collaboration for SFU is similar to that of MFU. The configuration of SFU is left for further study. The establishment of data channel for computing collaboration could be between SFU and MFU or between SFU and OLT.

5 Scenarios and Computation Capabilities

5.1 F5G Use Cases relevant to computing collaboration

Computing collaboration enhances QoE for digital services and enables more efficient network management. This capability applies to a number of F5G use cases [i.1]:

- **F5G R1 use case #1: Cloud Virtual Reality**

This use case describes the Cloud VR scenario, offloading computing and cloud rendering in VR services from local dedicated hardware to a shared cloud infrastructure. Cloud rendered video and audio outputs are encoded, compressed, and transmitted to user terminals through fast and stable networks.

In this use case, computing collaboration identifies VR service priority and optimizes network configuration to provide sufficient network capability for fulfilling service requirements in terms of throughput, latency and jitter.

- **F5G R1 use case #10: Scenario based broadband**

This use case describes the need for the network to adapt to service requirements from various coexisting frequently used Internet applications, such as education, gaming, media streaming, and ordinary best-effort Internet (browsing, and Email).

In this use case, computing collaboration identifies service types based on application characteristic. Network resources are then allocated to support corresponding network demands with adjustments made based on flexible changes of the broadband application in the Internet.

- **F5G R1 use case #11: Enhanced traffic monitoring and network control in intelligent access network**

This use case describes how dynamic data collection and analysis improves traffic monitoring capabilities in the optical access network enabling further management decisions for optical components.

In this use case, computing collaboration analyses traffic characteristics to fine-tune network parameters including both PON and FTTR networks.

- **F5G R1 use case #14: Digitalized ODN/FTTX**

This use case describes a method to digitalize ODN infrastructure through ODN labels and provides intelligent identification, achieving 100 % accurate resource recording and digital management enabling automatic service provision.

In this use case, computing collaboration provides visualization and management of fibre infrastructure, ensuring accurate identification and recoding of ODN resources thereby facilitating rapid ODN deployment and simplified maintenance.

- **F5G R2 use case #15: XR-based virtual presence**

This use case describes virtual presence based on XR, where users consume immersive content while producing immersive content for other online end users.

In this use case, the computing collaboration addresses the challenges of XR-based remote collaboration by offering a true feeling of being together achieved through provision of sufficient real-time bitrate and deterministic latency.

- **F5G R2 use case #18: Virtual music**

This use case describes sharing and creating music in real time by different parties that are not in the same location. The applications include virtual orchestra, virtual music classes, virtual concerts, etc. Currently, this is hardly possible. This require strict synchronization and latency requirements.

In this use case, the computing collaboration is used to enable a E2E deterministic latency.

- **F5G R2 use case #30: Bandwidth on demand**

This use case describes adaptive network planning based on the static throughput characteristic for the majority of the digital services, such as online gaming, periodic data backups, premium video streaming service, newspaper download, pre-recorded video download, etc.

In this use case, the computing collaboration helps service providers optimize available resources, develop new business propositions for end-users and improve the QoE for end-users.

- **F5G R2 use case #32: AI-based PON optical path diagnosis**

This use case describes identification of optical path fault information in the optical access network through data collection training, and analysis via an AI platform. The AI platform learns the fault characteristics and accurately models the ODN data with PON optical path data.

In this use case, the computing collaboration identifies the root cause of a PON optical path faults and enables rapid fault location. Optical path performance degradation is monitored and identified before service interruption actually occurs. The associated work order system automatically sends orders in advance for optical path fault repair preventing customer service interruption and significantly improving repair efficiency

5.2 F5G-A Use Cases relevant to computing collaboration

The F5G-A Use case #3 in ETSI GR F5G 020 [i.2] explores the collaboration of computing power in the PON access network. The computing power, located in ONT, OLT or cloud platform, is used for different purposes, but primarily to improve the service QoE. The computing power of ONT, OLT and cloud platform at each level are as follows:

- ONT: evaluates and analyses the status of the on-premises network based on its computing capabilities.
- OLT: The computing power within the OLT derives a deeper and more accurate view of on-premises network status based on ONT feedback, and assist the OLT in performing certain operations such as service identification, DBA, QoS, etc.
- Cloud platform: performs the deepest analysis of service quality, analysing the service over a long period of time, providing more accurate conclusions, and request other systems/platforms to collaboratively execute the necessary operations.

Furthermore, there are several additional related F5G-A use cases in ETSI GR F5G 020 [i.2] involving the computing power in the PON access network:

- **F5G-A R3 use case #2: Stable & reliable Wi-Fi® connection over FTTR**

This use case describes several scenarios (Gigabit/multi-Gigabit coverage, latency sensitive services, seamless hand-over for service continuity, and robust control for IoT services) requiring stable and reliable Wi-Fi® connections over FTTR.

In this use case, the computing power identifies Wi-Fi® problems and improve Wi-Fi® quality, e.g. service type identification and adaptation, and Wi-Fi® interference recognition and avoidance. In addition, the computing power facilitating seamless handover between different APs within a single FTTR network or across multiple FTTR networks.

- **F5G-A R3 use case #9: Evaluation and assurance of user service experience**

This use case describes proactive evaluation and assurance of user experience with fast service experience problem identification and rectification in the PON access network.

In this use case, computing power in ONT, OLT and QoE analysis system (which may reside in the cloud platform) executes algorithms for user experience evaluation, QoE problem root cause analysis and user experience rectification.

- **F5G-A R3 use case #11: Dynamically digitalized ODN**

This use case describes dynamically digitalized ODN to perform proactive and periodical ODN inspection.

With computing power in OLT (called Optical Artificial Intelligent) and the cloud platform, various physical parameters collected from passive intelligent optical components (e.g. splitters) serve as input for AI algorithms to perform fault prediction, fault detection and fault recovery in the ODN.

- **F5G-A R3 use case #12: On-premises Millimetre Wave (mmWave) WLAN**

This use case introduces the mmWave WLAN for high throughput (e.g. 10 Gbps) and high QoE light-of-sight connections, avoiding sub-6G Wi-Fi® interference. Sub-6G Wi-Fi® may still be used to assist beamforming and tracking of mmWave signals.

In this use case, the computing power executes algorithms for mmWave beamforming and tracking.

- **F5G-A R3 use case #17: QoD App-Flow service provisioning**

This use case introduces dynamic App-Flow service provisioning for the home broadband users, to improve the user experience.

In this use case, the computing power at both the NE-level (e.g. ONT and OLT) and the controller-level (which may run in the cloud platform) are aware of the user service intention (e.g. SLA), perform App-Flow prediction and service package development, and App-Flow SLA analysis.

In addition, F5G-A introduces new cloud services that requires high bandwidth and very strict latency, including two use cases in ETSI GR F5G 020 [i.2] that are listed below:

- F5G-A R3 use case #6: Naked-eye 3D display
- F5G-A R3 use case #10: Cloud Desktop

To improve network performance for these services, one possible solution is to move the cloud application content and the related servers (e.g. rendering servers) closer to users. This approach requires collaboration of computing power in the PON access network.

EXAMPLE: Edge Content Delivery Network (CDN) storage and Graphics Processing Unit (GPU) computing power for the cloud applications is installed on the OLT.

5.3 Additional use case: demonstration of applying computing-networking collaboration in Cloud-VR

Figure 5 shows an example demonstrating the computing-networking collaboration functions in a Cloud-VR service. In this case, the ONT discovers the on-boarding of a VR device. The peer OLT establishes the computing-networking collaboration channels for service characteristic identification and make decisions for network optimization. The OLT maintains the lifecycle of this computing-networking collaboration channel (including channel setup and deletion). The OLT simultaneously investigates the data demand of the VR service. Furthermore, this is aligned and collaborated with the transport network, like OTN to reserve a rigid data transmission channel from the rendering cloud.

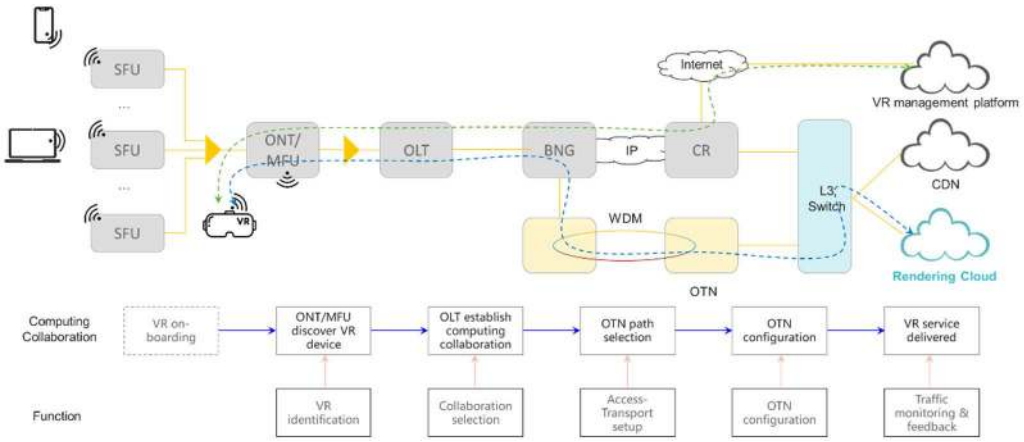


Figure 5: An example of applying computing collaboration in Cloud-VR scenario

6 Collaboration Requirements

6.1 Collaboration Scenarios

The collaboration scenarios include:

1) Collaboration between on-premises devices MFU and SFU.

2) Collaboration between PON devices OLT and ONTs/MFUs.

3) End-to-End collaboration across the optical access network between the PON network and the on-premises network.

6.2 General Technical Requirements for Computing-Networking Collaboration

The main characteristics of computing-networking collaboration are reflected in the use of agile, reliable, intelligent, and secure network facilities to connect the centre and edge computing facilities. This enables unified orchestration and control of multi-level computing resources, allowing computing power to be rapidly, securely, and intelligently transmitted over the network. This, in turn, enables computing power to assist the network in achieving dynamic, on-demand, and intelligent connections.

Specifically, it includes the following 4 dimensions of collaboration:

1) **Computing power collaboration:** computing power collaboration refers to the distribution of computing power resources on the access network and broadband home network based on the perception of user requirements and the distribution of computing power resources on the access network and broadband home network.

   Hierarchical scheduling and orchestration, select proper paths based on the orchestration result and routing policy, and schedule computing power requirements to corresponding nodes.

2) **Data collaboration:** data collaboration refers to the matching between data and computing power through the network, thereby forming collaboration between data distribution and computing.

   Data collaboration includes data collection, storage, processing, and analysis, and effective matching between data locations and computing power. In this way, effective collaboration between central offices and terminals, and between systems and devices at the management and control layer is realized.

3) **Intelligent collaboration:** intelligent collaboration schedules services to appropriate nodes based on the collaborative perception of services, computing resources, and network resources, implementing collaborative invoking of intelligent capabilities.

4) **Slice collaboration:** slice collaboration implements slice service provisioning and control by sensing, measuring, calculating, and controlling the transmission quality of network slices in real time, improving the assurance of slice transmission quality.

Table 1: Functional Requirements for Compute-Network Convergence by Collaboration Dimension

Dimension of Collaboration	Capability	Description	Orchestration Layer	M&C Layer	Device Layer (CO & Terminal)
Compute	**Measurement**	Unified modelling of multi-source heterogeneous compute (CPU, NPU, GPU). Unit: TOPS (int8), FLOPS (FP32).	Mandatory	Mandatory	Mandatory
	Awareness	Real-time discovery and reporting of available compute resources and their status.	Mandatory (aggregate)	Mandatory (collect from devices)	Mandatory (report local status)
	Scheduling	Assigning application compute requests (e.g. inference tasks) to the optimal node based on policy (latency, load, proximity).	Mandatory	N/A	N/A
	Management	Lifecycle management (provisioning, monitoring, decommissioning) of compute resources.	Mandatory (orchestration)	Mandatory (ops & maintenance)	Mandatory (local execution)
Data	**Collection**	Gathering operational data (state, attributes, KPI metrics) from devices and network paths.	N/A	Mandatory (from devices)	Mandatory (push to M&C)
	Storage	Persistent storage of telemetry, logs, and user-data cache (e.g. video). Supports configurable data retention (e.g. ≥ 30 days).	N/A	Mandatory (aggregated data)	Optional (local cache & logs)
	Processing	Hierarchical data handling. Edge devices perform real-time filtering; M&C layer performs correlation and analytics.	N/A	Mandatory (central analysis)	Optional (edge filtering)
	Transmission	Secure, lossless transport of critical operational data. TLS 1.3 encryption is mandatory. Zero-packet-loss is targeted for critical flows.	N/A	Mandatory (secure protocols)	Mandatory (secure protocols)
Intelligence	**Hierarchical Closed-loop**	Automated, localized response to events at each level without central intervention (e.g. ONT adjusts Wi-Fi®; OLT adjusts DBA).	Mandatory (global policy)	Mandatory (cross-device policy)	Mandatory (local automation)
	Cross-domain Collaboration	End-to-end coordination across Broadband Customer Network and Access Network for key user/services.	Mandatory (via northbound)	Mandatory (cross-domain control)	Mandatory (execute commands)
Slice	**Lifecycle Management**	On-demand creation, modification, and deletion of application-level slices (e.g. for gaming, VR).	Mandatory (orchestrate E2E)	Mandatory (provision per domain)	Mandatory (instantiate/apply)
	SLA Assurance	Real-time monitoring and deterministic guarantee of slice SLAs (latency, jitter, loss).	Mandatory (policy monitoring)	Mandatory (KPI collection)	Mandatory (measure and report)
	Closed-loop O&M	Automated "discovery -> provisioning -> intelligent O&M -> performance demonstration" for slice services.	Mandatory (policy engine)	Mandatory (analytics & root cause)	Mandatory (execution & reporting)

The collaboration scenarios include the collaboration between on-premises devices (i.e. MFU and SFU), collaboration between PON devices (i.e. OLT and ONT) and collaboration across the E2E optical access network between PON network and on-premises network:

1) **The collaboration between on-premises devices:**

   Requirements for enabling stable and reliable Wi-Fi® connections as described in F5G-A Use case Release 3 [i.2], the fibre-based on premises network shall support:

   1) Network application identification and corresponding QoS mechanisms;

2) An on-premises network centralized controller;

3) Real time Wi-Fi® channel information collection;

4) Continuous real time Wi-Fi® transmission behaviour configuration of each Wi-Fi® AP

5) A low latency controlling channel with a 64 microsecond round trip time between the optical transceivers of MFU and SFU

6) A control messaging procedure with bounded latency as required by the identified QoS

2) **The collaboration across the E2E optical access networks between PON network and in-premises network:**

As described in F5G-A Use case #3 [i.2], the general requirements are:

1) The PON network elements shall support additional computing-oriented hardware or modules;

2) The PON network shall support service flow identification and identification of corresponding QoS requirements;

3) The cloud platform shall support enabling or disabling computing collaboration of the OLT, MFU or SFU;

4) The cloud platform shall be capable of collecting computing capability information and utilization of the OLT, MFU or SFU computing resources

5) The OLT shall support a computing collaboration information and configuration exchange interface with the MFU and SFU, this interface shall be protected by security mechanisms.

6) The MFU in the on-premises network shall support a computing information and configuration exchange interface with the SFU, this interface shall be protected by secure mechanism.

Enabling centralized control over PON and on-premises network, as described in F5G-A Use case #7 [i.2]:

1) The OLT shall be capable of requesting and receiving transmission status (including data buffer, communication environment) from any MFU and SFU.

2) The OLT shall be capable of configuring the transmission behaviour of any MFU and SFU, including the optical transmission parameters in the optical access and on-premises network, and air interface parameters of the WLAN.

3) The MFUs shall be capable of providing local control for the SFU and corresponding end-user devices through a centralized control mechanism

Enabling centralized management over PON and on-premises network, as described in F5G-A Use case #7 [i.2]:

1) The optical access and on-premises network shall support an extension of the OMCI management interface.

2) The MFUs shall be capable of independently managing the SFUs, without invocation of the OLT or the management system.

3) The management system shall support management coordination between OLT and MFUs.

4) The unified management system shall support collection of device information from the different network elements once they are connected via optical fibre.

5) The unified management system shall provide visualization of the logical link for different service flow, to enables a clean mapping between service flow and scheduling strategy.

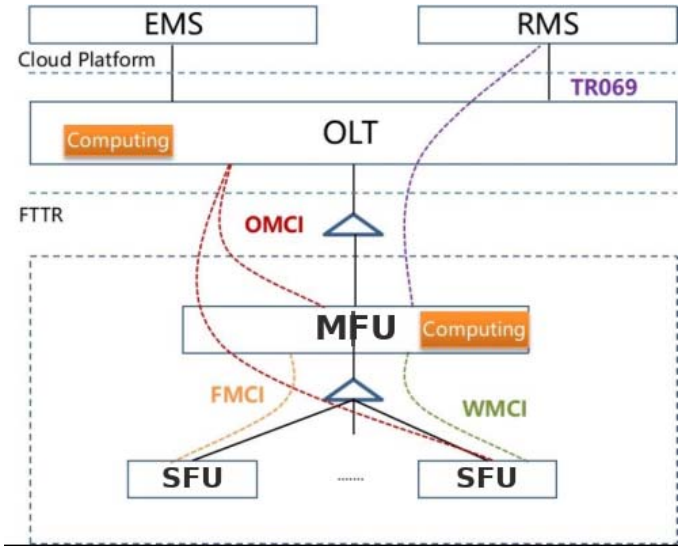
7 Components Enhancement

7.1 General

To facilitate computing collaboration functionalities, components of the PON and on-premises network shall be enhanced. These enhancements include:

- Enhancement on the heterogeneous computing capability, requiring additional and dedicated computing hardware and modules, integrated in the OLT, MFU and SFU);
- Enhancement on the perception capability. The PON network, including the OLT, MFU, SFU and the cloud platform, shall be capable of identifying the service flows and recognizing corresponding computing service demands;
- Enhancement on the dynamic computing collaboration in the PON network. The protocol and data model for controlling and managing computing capability includes OMCI, FMCI, WMCI, and BBF TR-069 [i.6].

The computing collaboration scheme in the PON network is shown in Figure 6.



NOTE: The terms E-ONU and P-ONU refer to SFU and MFU (different F5G-A release architecture use different naming).

Figure 6: The computing collaboration scheme in PON network

7.2 Enhancement on Heterogeneous computing capability

Table 2 below describes capabilities applicable to device-level and management-level layers (Perception, Computing, Connectivity, Storage, Control):

Table 2: Core Capabilities by Layer (Perception, Computing, Connectivity, Storage, Control)

Capability	Description	OLT	ONT/FTTR	Management & Control	Orchestration
Perception Capability					
Self-status awareness	Device CPU/memory/temperature/health monitoring.	Mandatory	Mandatory	Mandatory	
Device capability awareness	Supported features, firmware version, compute capacity (TOPS, cores, memory). Only Wi-Fi® line awareness (channels, SNR, interference) is required; PON line awareness is optional.	Mandatory	Mandatory	Mandatory	
Line capability awareness	PON: GEM ports, bandwidth, slicing, PON ports, WAN ports. Wi-Fi: channels, SNR, interference.	Optional	Mandatory (Wi-Fi only)	Mandatory	
Application-level flow identification	Classify flows by 5-tuple + application signature (VR, gaming, video, VoIP). Management layer shall aggregate application flow data from all OLTs and ONTs.	Mandatory	Mandatory	Mandatory	
Experience KPI monitoring	Per-flow latency, jitter, loss, throughput. Application identification shall support at minimum: VR, gaming, video streaming, VoIP.	Mandatory	Optional	Mandatory	
Computing Capability					
Network SLA calculation	Compute achievable SLA per user/application based on current resources.	Mandatory	Optional	Mandatory	
Device/line SLA requirements	DBA scheduling parameters, slicing requirements, queue depths.	Mandatory	N/A	Mandatory	
AI/ML model support	Inference (OLT/ONT) and training (Mgmt/Orch) of models. OLT inference latency shall be ≤ 50 ms (p99) per flow. ONT inference is recommended but not mandatory for low-end devices. This layer is responsible for model training and distribution, not real-time inference.	Mandatory (interference)	Mandatory (interference)	Mandatory (interference)	Mandatory (interference)
Connectivity Capability					
App-flow basic attributes	Include but not limited - define 5-tuple, latency requirement, deterministic class.	Mandatory	Mandatory	Mandatory	
App-flow lifecycle management	Create, modify, update, delete flows.	Mandatory	Mandatory	Mandatory	Mandatory (via policy)
App-flow to legacy mapping	Map app-flow to GEM port, VLAN, or tunnel.	Mandatory	Mandatory	Mandatory	
Storage Capability					
Operational data storage	Poor quality events, intermittent disconnections, anomaly logs. Management layer operational data retention shall be ≥ 30 days for analytics.	Mandatory	Mandatory	Optional	
User data storage	CDN cache (but not limited to), camera buffer, VR frame cache User data storage on OLT is implementation-dependent and optional.	Optional	Optional	N/A	

Capability	Description	OLT	ONT/FTTR	Management & Control	Orchestration
Storage Capability					
Event restoration	Replay any event within configurable window (e.g. 7 days).	Mandatory	Mandatory	Mandatory	
Control Capability					
Event detection from data	Detect SLA violations at the second level, quality degradation from stored patterns.	Mandatory	Optional	Mandatory	
Dynamic flow/network adjustment	Re-route or re-prioritize flows based on compute results. ONT adjustments limited to Wi-Fi parameters (channel, transmit power, airtime fairness).	Mandatory	Optional	Mandatory	Mandatory (via policy)
Resource allocation control	Allocate compute cores, memory, storage buffers.	Mandatory	N/A	Mandatory	Mandatory
Second-level SLA monitoring	Detect and report SLA breaches within ≤ 1 second.	Mandatory	Optional	Mandatory	

7.3 Enhancement on computing functionality (including perception capability)

Table 3 describes capabilities specific to the Orchestration Layer.

Table 3: Collaborative orchestrating System Functionality (Cross-Domain Coordination)

Multi-source compute orchestration	Unified model for OLT, ONT, edge cloud compute. Compute measurement unit: TOPS (int8) for inference, FLOPS (FP32) for training	Mandatory
Joint compute + network orchestration	Balance compute scheduling and network path planning.	Mandatory
Dynamic compute re-orchestration	Adjust compute allocation based on load changes.	Mandatory
Data collection orchestration	Coordinate data aggregation from OLT, ONT, Management.	Mandatory
Data storage & processing workflow	Hierarchical processing based on data attributes.	Mandatory
Security policy orchestration	Encryption, integrity, path planning. Security policy orchestration includes mTLS, RBAC, and encryption-at-rest policies.	Mandatory
Cross-domain intelligent orchestration	Key service assurance across network domains.	Mandatory
Application-level slice orchestration	Automated provisioning, deployment, distribution of slices.	Mandatory
Slice SLA assurance	Orchestrate SLA monitoring and enforcement.	Mandatory
Multi-dimensional strategy coordination	Compute + data + intelligence + slice unified orchestration.	Mandatory
Conflict detection	Detect and resolve conflicting orchestration policies. Conflict detection shall handle ≥ 100 concurrent orchestration policies.	Mandatory

7.4 Enhancement on dynamic computing collaboration in PON network

7.4.1 Overview

To support dynamic computing collaboration, the management and control interfaces of the PON network and the on-premises network need to be enhanced. The present clause specifies the enhancement in terms of generic collaboration interfaces, denoted I1 to I5, and of the functions to be implemented over them. These generic interfaces are realized by extending the existing management and control protocols of the PON and the on-premises network.

EXAMPLE: Example implementation choices can be the following: the management interface between the OLT and the ONT might be OMCI [i.3], the in-premises management interfaces G.fin management [i.4], [i.5], the remote management protocols between the cloud platform and the customer-premises equipment might be CWMP [i.6], [i.7], [i.8], and the YANG-based northbound interface [i.9], [i.10].

For each interface, the functions, the generic mechanism and procedure, the message format and the data model are described.

The enhancement follows two complementary approaches:

a) extending the existing management and control protocols with computing-collaboration-specific data models, so that the procedures defined in clause 4.2.2 are realized; and

b) establishing an IP-based computing collaboration data channel for high-frequency telemetry data exchange and AI model distribution, when the capacity or the latency of the management channel is not sufficient.

7.4.2 Generic interface framework for computing collaboration

Figure 7 shows the generic interface framework for dynamic computing collaboration. Five interfaces are defined according to their connectivity and roles:

- I1: the access collaboration interface, between the OLT and the ONT/MFU and the SFU forwarded by the MFU;
- I2: the in-premises collaboration interface, between the ONT/MFU and the in-premises nodes (SFUs);
- I3: the remote management interface, between the NMS and the ONT/MFU;
- I4: the northbound interface, between the NMS and the OLT;
- I5: the computing collaboration data channel, an IP-based logical channel between the OLT and the ONT/MFU and the SFU forwarded by the MFU.

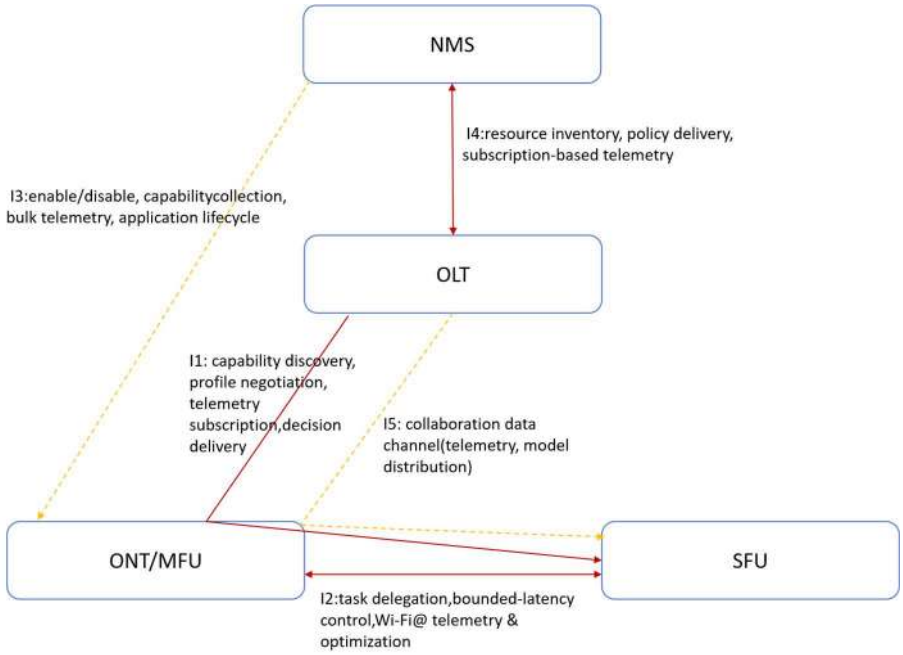


Figure 7: Generic interface framework for dynamic computing collaboration in a PON network

Table 4 summarizes the functions to be implemented over each interface. The detailed protocol stack realizing each interface is implementation-specific and is left to the protocol data model specifications.

Table 4: Functions to be implemented over the collaboration interfaces

Interface	Connectivity	Functions to be implemented
I1	OLT ↔ ONT/MFU and SFU	Discovery and reporting of computing capabilities; negotiation of collaboration profiles; subscription and reporting of telemetry data; delivery of collaboration decisions and configuration; lifecycle management of on-device analytics models; application-flow awareness.
I2	ONT/MFU ↔ SFU	Advertisement of in-premises computing capabilities; delegation of computing tasks and collection of processing results; bounded-latency control messaging; Wi-Fi® telemetry reporting and optimization delivery.
I3	NMS ↔ ONT/MFU	Remote enabling/disabling of the collaboration function; collection of computing capability and utilization; bulk data collection for AI/ML training and analytics; lifecycle management of downloadable computing applications.
I4	NMS ↔ OLT	Delivery of orchestration policies; subscription-based real-time telemetry reporting; exposure of collaboration capabilities as network services.
I5	OLT ↔ ONT/MFU and SFU (IP-based)	High-frequency telemetry transfer; distribution of AI model packages and other bulk data; channel negotiated through the management interface (I1).

The functions listed in Table 4 shall be supported by extending the existing management and control protocols. The extensions shall be backward compatible: a network element that does not support computing collaboration shall ignore the extensions without impact on legacy management functions.

The I5 data channel shall be established when the management interface is not able to fulfil the bandwidth or latency requirements of the telemetry data exchange; its setup, maintenance and release shall be negotiated through the I1 interface.

The semantics of the computing collaboration information shall be consistent across the interfaces I1 to I4, so that the same information has the same meaning and representation on every interface.

All the interfaces shall support security functions including mutual authentication, integrity protection and encryption, as required for the collaboration information and configuration exchange in clause 6.

7.4.3 Enhancement on the access collaboration interface

The access collaboration interface (I1) is realized by extending the existing management interface between the OLT and the ONT, which operates in a master-slave manner over a dedicated management channel. To support computing collaboration, additional managed entities shall be introduced in the management information base, providing the capabilities listed in Table 5. The detailed definition of the managed entities, including their identifiers, attributes and actions, is left to the corresponding protocol specification and are for further study.

Table 5: Required new managed entities and the capabilities to be implemented

No.	Required new managed entity	Capabilities to be implemented
1	Computing resource reporting entity	Exposes the type, capacity and real-time utilization of the local computing resources (processing, memory, storage) of the network element, with unified measurement units (TOPS in int8, FLOPS in FP32), enabling the OLT to master the computing power of the ONT/MFU and the SFU.
2	Collaboration negotiation entity	Advertises the supported collaboration functions (e.g. data collection, data filtering, inference, caching), negotiates the collaboration role and profile between the OLT and the ONT/MFU, and enables or disables the collaboration function.
3	Data channel management entity	Establishes, maintains and releases the computing collaboration data channel (I5), binds the channel to the appropriate transport resources and to a security profile, and monitors the channel state.
4	Telemetry subscription entity	Subscribes KPI data required for collaboration decision-making with configurable reporting interval (down to 1 s for SLA-related KPIs) and reporting mode (periodic, on-change, on-threshold), and reports events autonomously to the OLT.
5	Analytics model management entity	Manages the lifecycle of on-device analytics and AI models, including model distribution, activation, version management and state monitoring, supporting collaborative inference between the OLT and the ONT/MFU (and the SFU).
6	Application-flow awareness entity	Configures application-level flow classification and reports identification statistics, providing the perception basis for collaboration decisions.

For the message format, the following requirements apply:

- the existing message structure of the management interface shall be reused, and the new managed entities shall be discoverable through the management information base discovery mechanism, so that legacy implementations remain unaffected;
- an extended message format shall be supported for the transfer of attributes whose length exceeds the capacity of the baseline message, including bulk telemetry data; and
- an autonomous notification mechanism shall be supported to report collaboration events, e.g. threshold crossing of subscribed KPIs and collaboration state changes.

For the generic mechanism and procedure, the interface shall support the general procedure defined in clause 4.2.2, capability discovery, profile negotiation, data channel setup, data subscription, notification, decision delivery and outcome reporting, realized through the capabilities in Table 5.

Telemetry reporting with an interval down to 1s shall be supported for SLA-related KPIs, fulfilling the second-level SLA monitoring requirement in clause 7.2. The computing collaboration traffic over the management channel shall be prioritized and rate-limited such that legacy management functions are not degraded. Transfer of large data volumes, should use the data channel (I5) instead of the management channel.

EXAMPLE: An example for large data volumes is the transfer of AI model packages.

7.4.4 Enhancement on the in-premises collaboration interfaces

The in-premises collaboration interfaces (I2) are realized by extending the existing in-premises management interfaces.

EXAMPLE: Implementations might use G.fin management protocols [i.4], [i.5].

To support computing collaboration within the on-premises network, the following functions shall be supported:

- computing capability advertisement: an SFU or an AP reports its computing capabilities to the ONT/MFU, consistent with the computing resource reporting capability of the I1 interface;
- collaborative task delegation: the ONT/MFU delegates computing tasks (e.g. data filtering, Wi-Fi® analytics, inference) to the SFU and collects the processing results;
- bounded-latency control messaging: control messages are exchanged over a low-latency control channel with a 64-microsecond round-trip time between the optical transceivers of the ONT/MFU and the SFU, as required in clause 6, with the message priority configurable according to the identified QoS through the application-level flow identification (see clauses 6 and 7.2);
- Wi-Fi® telemetry reporting: Wi-Fi® KPIs (channel utilization, signal quality, interference, airtime) are reported to support collaborative Wi-Fi® optimization algorithms;
- optimization delivery: optimization actions derived by analytics, including channel selection, transmit power and airtime fairness as identified in clause 7.2, are delivered to the Wi-Fi® AP;
- assistance information exchange: assistance information for mmWave beamforming and tracking (F5G-A use case #12 [i.2]) and for seamless handover coordination (F5G-A use case #2 [i.2]) should be exchanged; and
- security: the secure channel and key management mechanisms of the underlying in-premises interfaces are reused.

7.4.5 Enhancement on the remote management and northbound interfaces

Over the remote management interface (I3), the following functions shall be supported:

- computing collaboration data model: data model objects covering computing capability, resource utilization, and the enabling/disabling of the collaboration function, as required for the cloud platform in clause 6; the ONT/MFU maps these objects to the data models of the I1 and I2 interfaces, so that the cloud platform masters the computing power resources and user requirements within the CPN as required in clause 4.1;
- bulk telemetry and eventing: large-scale data collection serving AI/ML training and analytics at the cloud platform;
- streaming transport: message transport supporting high-frequency streaming telemetry; and
- application lifecycle management: lifecycle management of downloadable computing applications on the ONT/MFU and the SFU.

Over the northbound interface (I4), the following functions shall be supported:

- unified modelling: a unified data model covering computing resource inventory and collaboration service provisioning;
- policy delivery: delivery of orchestration policies from the orchestration layer (clause 7.3) to the OLT;
- real-time telemetry: subscription-based real-time KPI reporting towards the MCA plane; and
- service exposure: exposure of the computing collaboration capability as part of NaaS with QoD, supporting real-time dynamic scheduling of computing power as required in clause 4.1.

The interfaces specified in the present clause shall support security functions including mutual authentication, role-based access control and encryption, consistent with the security policy orchestration in clause 7.3.

7.4.6 Generic mechanism, message format and data model requirements

Independently of the protocol realization, each collaboration interface shall support the following generic mechanisms: function enabling/disabling, capability discovery, profile negotiation, data channel lifecycle management (setup, keep-alive, release), data subscription and notification, decision delivery, and outcome reporting, in line with clause 4.2.2.

For the message format, the following generic requirements apply:

- extensibility with version negotiation and backward-compatible extension;
- transaction identification and error reporting with standardized result codes;
- message priority and bounded delivery latency for control messages, e.g. the low-latency control channel within the customer-premises network (clause 7.4.4); and
- fragmentation and bulk transfer support for telemetry and AI model data.

For the data model, the following generic requirements apply:

- a unified computing information model shall be applied across the data models of the interfaces I1 to I4, with consistent semantics and measurement units (TOPS in int8 for inference, FLOPS in FP32 for training, per clause 6);
- the data model shall cover the four collaboration dimensions defined in clause 6 (computing power, data, intelligence and slice);
- configurable data retention and privacy protection shall be supported, including data minimization and anonymization of user data; and
- data model versioning shall be supported to allow independent evolution of the protocol realizations.

NOTE: The detailed definition of the protocol data models realizing the interfaces I1 to I5 is for further study, in coordination with the relevant standards development organizations ([i.3] to [i.10]).

8 Conclusions

The present document specifies the F5G-A architecture of computing collaboration in the PON network. The architecture is characterized by distributed computing power across the NMP, the OLT, ONT and the FTTR devices, organized into an orchestration layer, a management and control layer, a computing layer and a forwarding layer, together with a general procedure of computing coordination covering function enabling, capability discovery, data channel setup, data subscription, decision delivery and outcome reporting.

Relevant F5G and F5G-A use cases have been analysed, and the collaboration requirements have been specified along four dimensions: computing power collaboration, data collaboration, intelligent collaboration and slice collaboration.

To fulfil these requirements, the enhancements of the PON network components have been specified in three aspects: the heterogeneous computing capability of the OLT, ONT and the FTTR; the computing functionality including perception capability, decomposed onto the network elements and the orchestration system; and the dynamic computing collaboration interfaces, specified in terms of generic collaboration interfaces and the functions to be implemented over them. These specifications summarize the capability gaps and will guide the future development of device specifications and multi-vendor interoperability.

The following items are for further study: the detailed definition of the protocol data models realizing the collaboration interfaces, to be progressed in coordination with the relevant standards development organizations; the security mechanisms of the collaboration interfaces; and the quantification of collaboration KPIs.

Annex A (informative): Change history

Date	Version	Information about changes
11/09/2023	V0.0.1	Propose table of content for WI-025
15/09/2023	V0.0.2	To reflect the comment received in the Sept plenary in 2023
29/05/2024	V0.0.3	Propose clauses 4.1 and 5.1 for WI-025
12/11/2024.	V0.0.4	The output from ISG F5G #21 plenary
04/02/2025	V0.0.5	Include the accepted contributions from #21 plenary
26/02/2025	V0.0.6	Include abbreviations and Editorial changes
31/07/2026	V0.0.7	Include the accepted contributions from F5G-CC_200
14/08/2026	V0.0.8	Include the accepted contributions from F5G-CC_201
19/08/2026	V0.0.9	Solve the comments from experts feedback and editor' note
26/08/2026	V0.0.10	Include the agreed modification from F5G-F2F#27

History

Version	Date	Status
V1.1.1	September 2026	Publication